

## Project Design Phase-II Technology Stack (Architecture & Stack)

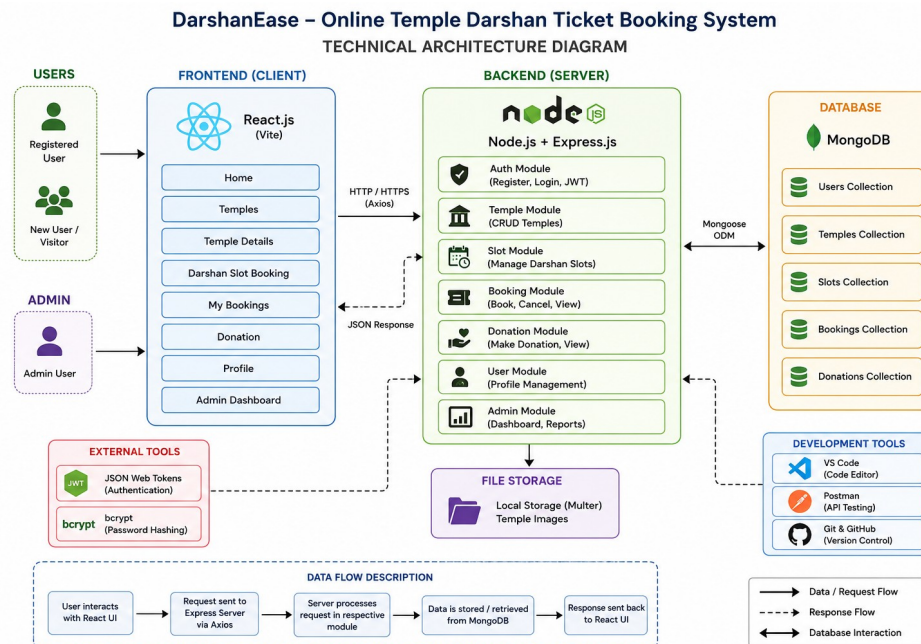
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|---------------|--------------|
| Date          | 14 July 2026 |
| Team ID       |              |
| Project Name  | DHARSHANEASE |
| Maximum Marks | 4 Marks      |

### Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

**Example: Order processing during pandemics for offline mode**

**Reference:** <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



**Table-1 : Components & Technologies:**

| S.No | Component                       | Description                                          | Technology                        |
|------|---------------------------------|------------------------------------------------------|-----------------------------------|
| 1.   | User Interface                  | Web interface for users and admin                    | React.js, HTML5, CSS3, JavaScript |
| 2.   | Application Logic-1             | Authentication and User Management                   | Node.js, Express.js               |
| 3.   | Application Logic-2             | Temple, Slot, Booking & Donation Management          | Express.js Controllers            |
| 4.   | Application Logic-3             | REST API Communication                               | Axios                             |
| 5.   | Database                        | Stores users, temples, bookings, slots and donations | MongoDB + Mongoose                |
| 6.   | Cloud Database                  | Not used (Local MongoDB)                             | N/A                               |
| 7.   | File Storage                    | Temple Image Upload                                  | Multer + Local Uploads Folder     |
| 8.   | External API-1                  | No external API used                                 | N/A                               |
| 9.   | External API-2                  | No external API used                                 | N/A                               |
| 10.  | Machine Learning Model          | Not applicable                                       | N/A                               |
| 11.  | Infrastructure (Server / Cloud) | Frontend, Backend and MongoDB running locally        | Ubuntu, Node.js, MongoDB          |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                                                      | Technology                             |
|------|--------------------------|----------------------------------------------------------------------------------|----------------------------------------|
| 1.   | Open-Source Frameworks   | Frontend and Backend developed using open-source frameworks                      | React.js, Node.js, Express.js, MongoDB |
| 2.   | Security Implementations | JWT Authentication, bcrypt password hashing, Protected Routes, Role-based Access | JWT, bcryptjs                          |
| 3.   | Scalable Architecture    | Three-tier architecture separating UI, Business Logic and Database               | MERN Stack                             |
| 4.   | Availability             | Runs continuously while frontend, backend and MongoDB services are active        | Node.js, MongoDB                       |
| 5.   | Performance              | Fast API communication using Axios and optimized MongoDB queries                 | Axios, Express.js, MongoDB             |

**References:**

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>